SESSION REPORT

Session d94c9109

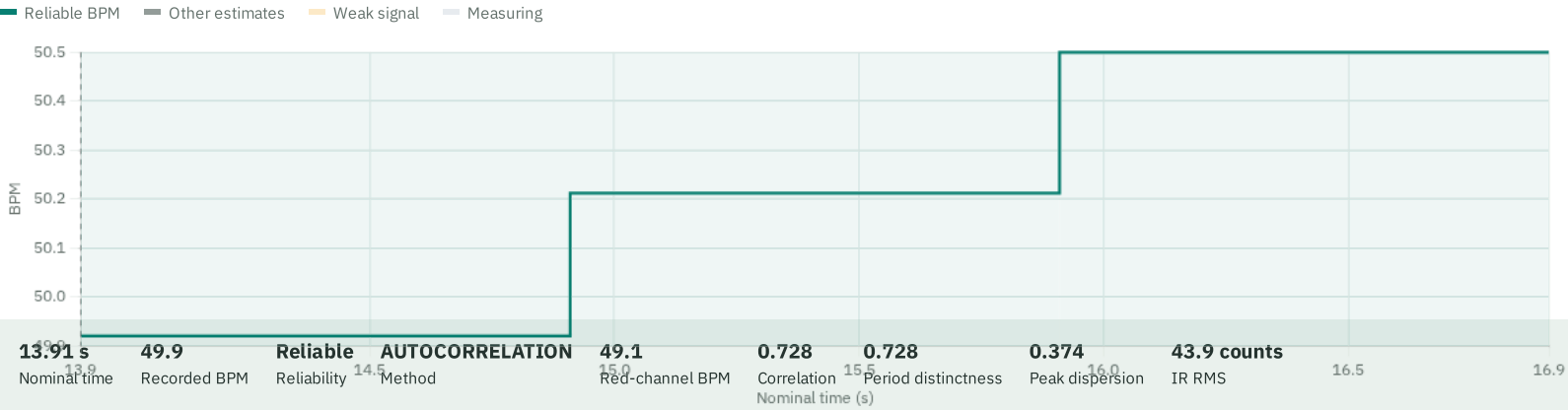
10/13/2026, 8:26:48 AM · 145.16 s · jc4880p433-heart

48.1% reliable coverage (69.78 s of 145.16 s). Summary statistics describe reliable periods only. Other estimates remain available for inspection.

Reliable mean	Reliable range	Reliable coverage	First usable reading	Detected peaks
47.2 BPM	44.9–53.1 BPM	48.1 %	13.91 s	123
Nominal-duration weighted	Reliable periods only	Not whole-session confidence	From recording start	Not verified heartbeats

Heart rate

13.9–16.9 s



13.91 s	49.9	Reliable	AUTOCORRELATION	49.1	0.728	0.728	0.374	43.9 counts
Nominal time	Recorded BPM	Reliability	Method	Red-channel BPM	Correlation	Period distinctness	Peak dispersion	IR RMS

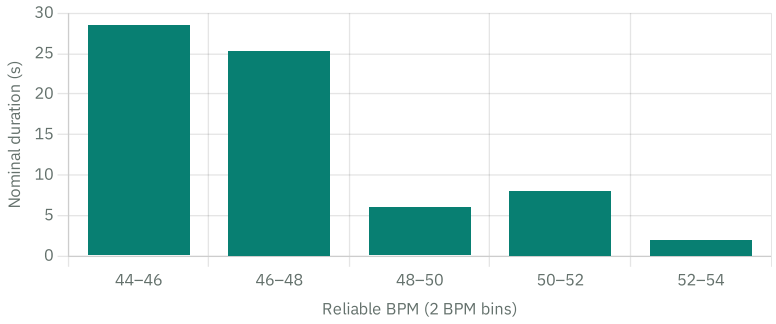
Flags: rate disagreement, unstable intervals · Reading at 13.91 s · Estimator window 3.04–13.04 s

Reliable periods

Recorded policy, not a clinical assessment

Period	Start	End	Duration
1	13.91 s	16.91 s	3.00 s
2	30.91 s	31.91 s	1.00 s
3	45.91 s	63.91 s	18.00 s
4	64.91 s	82.37 s	17.46 s
5	114.84 s	145.16 s	30.32 s

Reading distribution

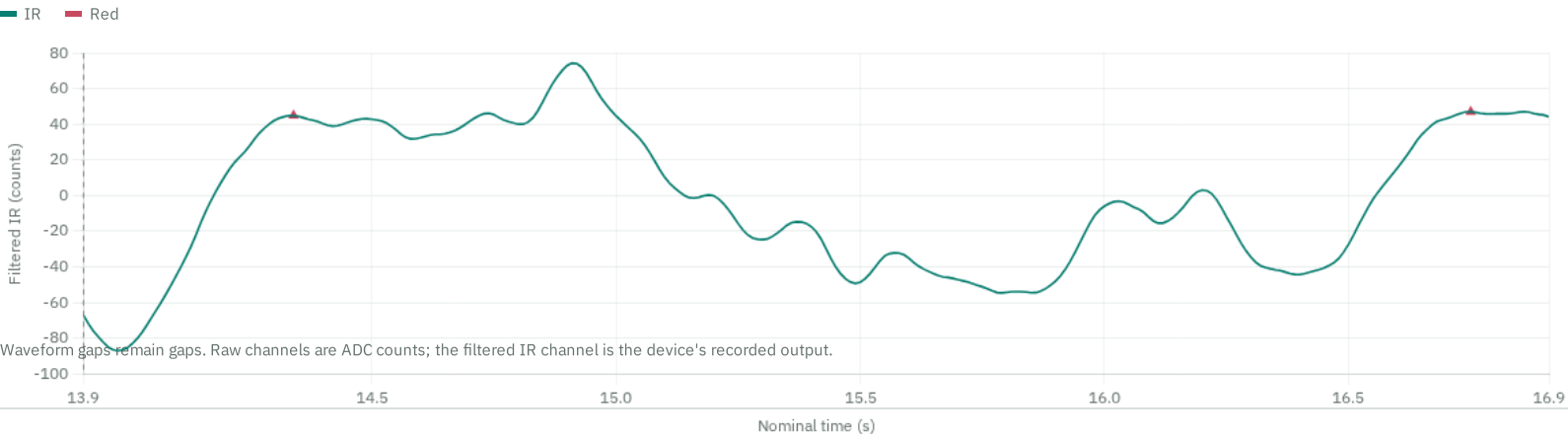


Capture at a glance

Reliable median	46.4 BPM
Usable signal	86.78 s (59.8%)
Longest reliable period	30.32 s
Stored samples	14,516
Lost samples	0
FIFO overflow counter	0
Capture status	Complete

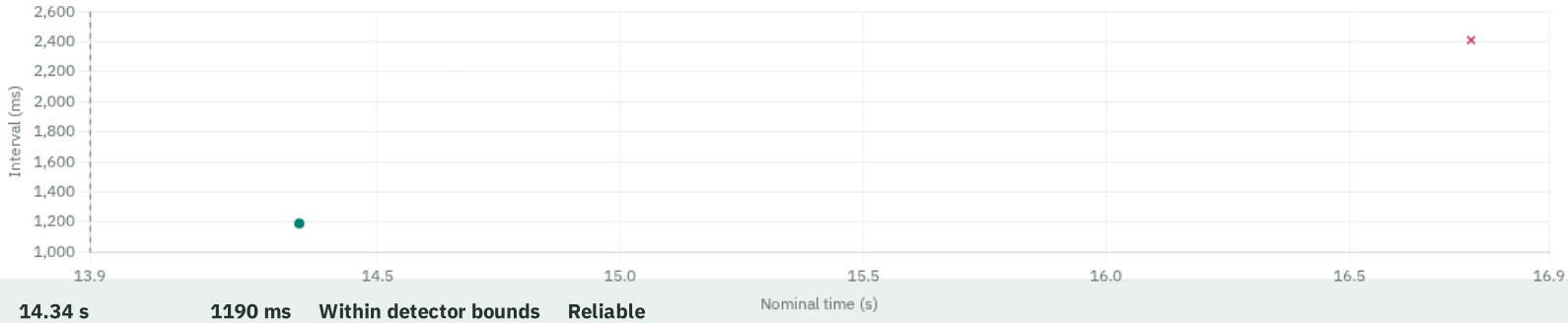
Optical pulse estimates are not ECG measurements. Detected peaks and in-range intervals are not independently verified heartbeats. No HRV or oxygen-saturation result is inferred.

Optical signals



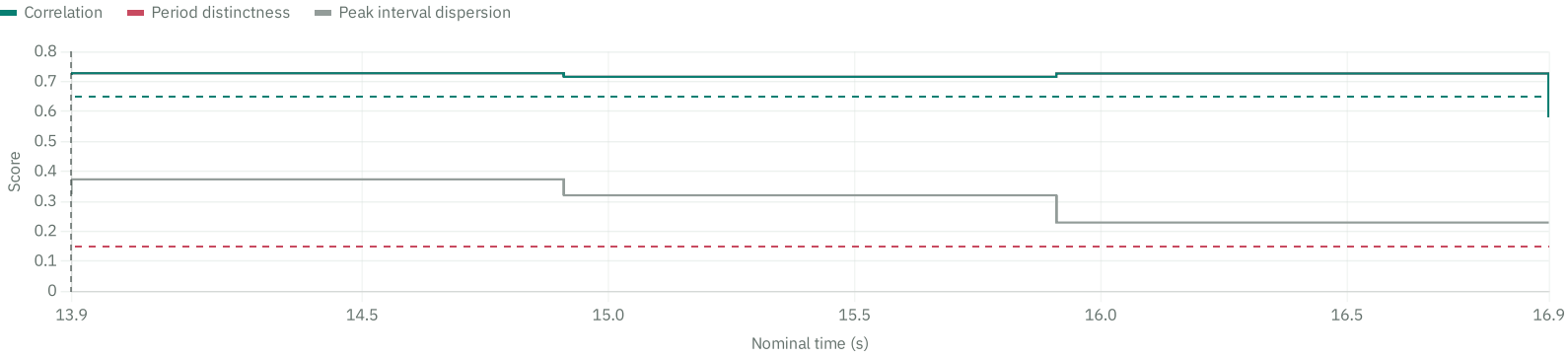
Waveform gaps remain gaps. Raw channels are ADC counts; the filtered IR channel is the device's recorded output.

Detected peak intervals



14.34 s	1190 ms	Within detector bounds	Reliable
Nearest detected peak	Interval	Interval check	BPM at detection

Estimator evidence



Session events

Time	Event	Status	Quality flags
	recording start		
	quality change	measuring	settling
	quality change	weak signal	abrupt change
	quality change	measuring	settling
	quality change	measuring	settling
	quality change	weak signal	rate disagreement, unstable intervals
	quality change	Reliable	rate disagreement, unstable intervals
	quality change	Reliable	unstable intervals
	quality change	usable	rate disagreement, unstable intervals
	quality change	usable	unstable intervals
	quality change	weak signal	unstable intervals
	quality change	weak signal	rate disagreement, unstable intervals
	quality change	weak signal	low periodicity, rate disagreement, unstable intervals
	quality change	weak signal	rate disagreement, unstable intervals
	quality change	weak signal	low periodicity, rate disagreement, unstable intervals
	quality change	Reliable	rate disagreement, unstable intervals
	quality change	weak signal	rate disagreement, unstable intervals
	quality change	usable	rate disagreement, unstable intervals
	quality change	weak signal	low periodicity, rate disagreement, unstable intervals
	quality change	weak signal	low periodicity, rate disagreement, unstable intervals, stale beat
	quality change	weak signal	low periodicity, unstable intervals
	quality change	weak signal	low periodicity, rate disagreement, unstable intervals
	quality change	weak signal	rate disagreement, unstable intervals
	quality change	weak signal	low periodicity, rate disagreement, unstable intervals
	quality change	weak signal	unstable intervals
	quality change	Reliable	unstable intervals
	quality change	Reliable	None
	quality change	Reliable	unstable intervals
	quality change	usable	unstable intervals, stale beat
	quality change	Reliable	unstable intervals
	quality change	Reliable	None
	quality change	Reliable	unstable intervals
	quality change	weak signal	abrupt change
	quality change	weak signal	no contact
	quality change	measuring	settling
	quality change	weak signal	abrupt change
	quality change	measuring	settling
	quality change	weak signal	abrupt change

Time	Event	Status	Quality flags
	quality change	measuring	settling
	quality change	measuring	settling
	quality change	weak signal	rate disagreement, unstable intervals
	quality change	weak signal	low periodicity, rate disagreement, unstable intervals
	quality change	usable	rate disagreement, unstable intervals
	quality change	Reliable	rate disagreement, unstable intervals
	quality change	Reliable	unstable intervals
	quality change	Reliable	None
	quality change	Reliable	unstable intervals
	quality change	Reliable	rate disagreement, unstable intervals
	quality change	Reliable	unstable intervals
	recording stop		

Acquisition

sample rate hz	100
sensor sample rate hz	100
sample average	1
time basis	stored_sample_index_not_hardware_sequence
read elapsed us basis	host_fifo_poll_not_sample_timestamp
sensor	MAX30101
sensor address	87
part id	21
revision	6
pulse width us	411
adc range na	4096
led red code	31
led ir code	31
fifo rollover	false
max poll gap us	12678
fifo overflow counter	0
ambient overflow events	0
lost samples	0
warnings	None

Estimator & reliability policy

version	heart_bpm_v1
window s	10
startup s	2
update s	1
min bpm	30
max bpm	180
highpass hz	0.5
lowpass hz	4
min correlation	0.35
min rms counts	2
refractory s	0.3
min contact counts	5000
waveform	causal_second_order_highpass_then_lowpass_ir_counts
Policy: version	heart_reliable_v1
Policy: requires complete capture	true
Policy: min correlation	0.65
Policy: min period distinctness	0.15
Policy: max red rate relative difference	0.15
Policy: max peak interval dispersion	0.15
Policy: minimum summary duration s	10
Policy: allows autocorrelation with unstable peak intervals	true
Policy: excluded quality flags	settling, no contact, clipped, flatline, abrupt change, low amplitude, low periodicity, ambiguous period, red disagreement, stale beat, out of range

Time basis & integrity

CSV pairing	Verified: sample count, contiguous indices, byte size and CRC32
CRC32	9bc0b2b4
Nominal duration	145.16 s
Host duration	144.625 s
Maximum host polling gap	12.678 ms
Firmware	1.40.0
Stop reason	manual stop
Exercise stop marker	Not recorded
Annotations	[]

The horizontal axis uses stored sample index / effective sample rate. FIFO read elapsed time is host polling time, not a hardware sample timestamp. The report does not repair or re-estimate the capture.